AudioConditioner

Applications

CSC_52002_EP - Multimodal generative AI

APPLICATION I: IMAGE TO AMBIENT MUSIC

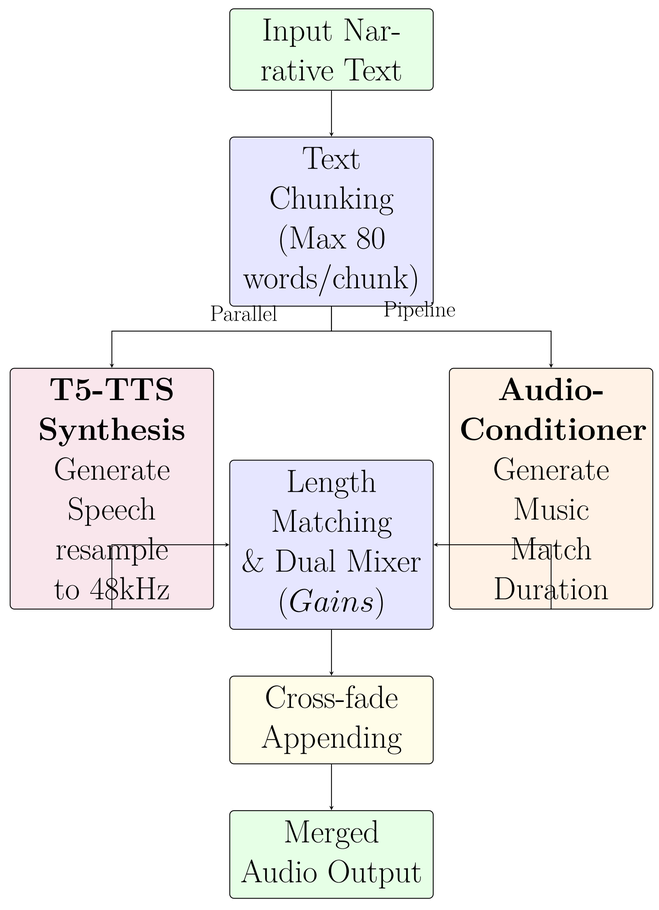
One direct and powerful application of the **AudioConditioner** framework is generating ambient music and atmospheres based solely on visual inputs. By passing an image into our pipeline:

- The **BLIP Model** automatically captions the image's scene, extracting emotional elements, colors, objects, and implicit narrative.
- This scene description drives the **AudioConditioner** pipeline (through the **CLAP Descriptor** and **Stable Audio**), outputting high-fidelity audio that perfectly encapsulates the mood of the image.
- The audio candidates are re-ranked using the **CLAP Cosine Similarity Filter** to ensure the highest alignment with the initial visual.

This enables seamless soundtrack generation for static artwork, game environments, or visual media.

APPLICATION II: TEXT READING WITH ADAPTIVE MUSIC

The **ReadMusic** module merges Speech Synthesis with adaptive background music scoring, turning textual narratives into fully immersive audiobooks.



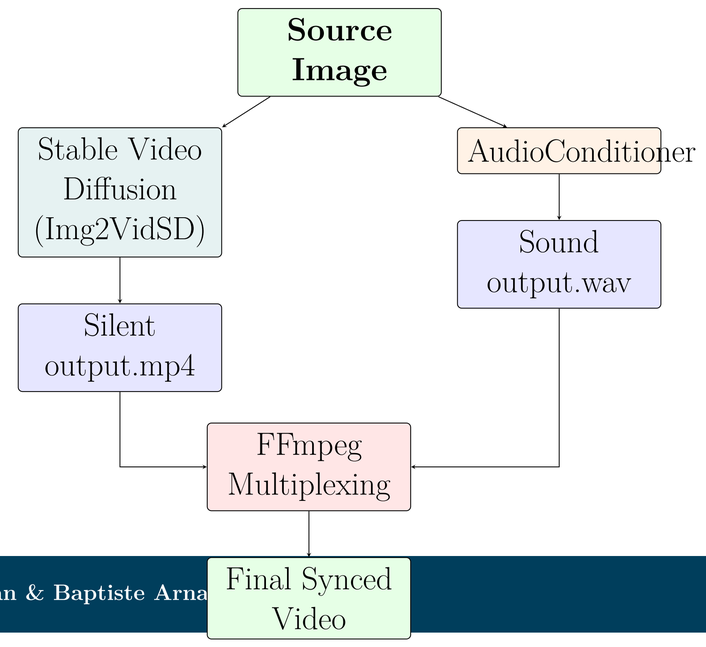
The text is robustly split using natural pauses (punctuation, newlines). For each chunk — mandated due to the strict **47-second generation maximum** of the under-the-hood Stable Audio model —, the script runs the T5 text-to-speech engine concurrently with our Music Generation model. Finally, speech and music arrays are zero-padded to match lengths, multiplied by their specific gains, and blended with dynamic crossfades to avoid clipping artifacts.

APPLICATION III: VIDEO GENERATION & ORCHESTRATION

Taking the visual modality further, the `video_sound.py` application creates animated cinematic clips with a tightly synchronized and responsive musical score from a single still image.

Parallel Execution Pathway:

1. **Image Animation:** The input image is parsed through the **Img2VidSD** Model (Stable Video Diffusion pipeline). This produces a smooth, non-static silent video sequence.
2. **Image Orchestration:** Simultaneously, the image goes directly into the **AudioConditioner** model to extract visual semantics and produce a matching audio waveform.
3. **Multimodal Merging:** We integrate **FFmpeg** through subprocesses to automatically compile the generated silent video (`.mp4`) and the generated audio (`.wav`), multiplexing them into a single presentation-ready file.



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These applications showcase the great versatility of **AudioConditioner**, positioning it as a strong foundation for automated scoring, procedural art generation, and immersive multimodal storytelling.

IF WE HAD MORE TIME: PPO FINE-TUNING

Given more time and compute budget, we would have fine-tuned our model using **Proximal Policy Optimization (PPO)**. By using the CLAP similarity score (between the generated audio and the text prompt) as a reward signal, the model could strictly optimize itself to directly maximize alignment instead of just distilling a Teacher LLM.

- **The Bottleneck:** In RL algorithms like PPO, the agent needs to continually generate outputs (rollouts) to explore the environment and receive rewards.
- **Order of Magnitude:** A single audio generation step with **Stable Audio** takes approximately **1 minute** to complete.
- **Conclusion:** Gathering enough generation trajectories to meaningfully update the network via PPO would thus require **weeks of continuous computing time**, making it unfeasible within the scope of this project.

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